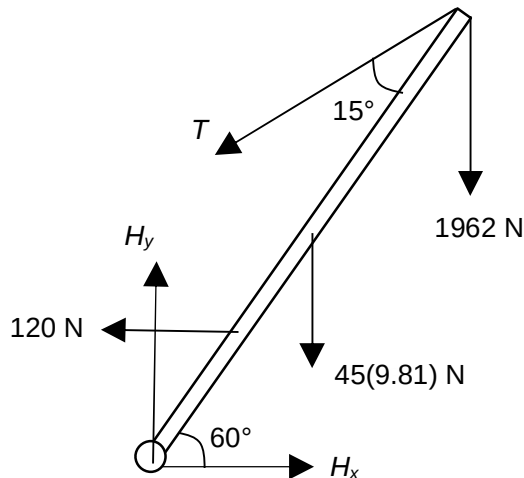


- 1 (a) Net force is zero (in all direction). [1]
Net moment is zero about any point. [1]
DO NOT ACCEPT: ...a point OR a given point

(b)



Evidence of angle between T and boom (in working) = 15° [1]

Taking moments about the hinge,

$$(T \sin 15^\circ)(3) + (120 \sin 60^\circ)(0.75) = (45)(9.81)(1.5 \cos 60^\circ) + (1962)(3 \cos 60^\circ)$$

[1 for correct CW moment]
 [1 for correct ACW moment, accept ecf 15°]

$$T = 4116 = 4.1 \text{ kN}$$

- (c) (Let H_y and H_x be the vertical and horizontal components of the force hinge acts on the boom)

$$H_y = 1962 + 45(9.81) + T \cos 45^\circ = 5314 \text{ N} \quad [1]$$

$$H_x = 120 + T \sin 45^\circ = 3030 \text{ N} \quad [1]$$

$$\text{Therefore, magnitude of force hinge acts on boom} = \sqrt{5314^2 + 3030^2} = 6120 \text{ N}$$

[1, ECF]

$$\text{Direction of the force} = \tan^{-1} (5314 / 3030) = 60.3^\circ \text{ above horizontal} \quad [1, \text{ECF}]$$